

Back to Square One

Foundations of AI

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How this works

- Nineteen questions, four answers each, and **no devices**.
- I read the question, you put a hand up for A, B, C or D.
- Then I click once and the answer appears.
- Everything on these slides was on a slide today.

A question the room gets wrong is worth five it gets right, so guess out loud and be wrong cheerfully. That has been the whole day.

1

Section 1

The loop

1. The line that does the learning

This is this morning's algorithm, finding a square root. One line is missing.

```
def root(area, alpha, tol, max_steps):  
    w = 1.0  
    l = 1.0  
    steps = 0  
    while abs(l) > tol and steps < max_steps:  
        l = area - w * w  
        ???  
        steps = steps + 1  
    return w
```

- A) `w = w + alpha * l`
- B) `w = w - alpha * l`
- C) `w = alpha * l`
- D) `w = w + alpha / l`

1. The line that does the learning

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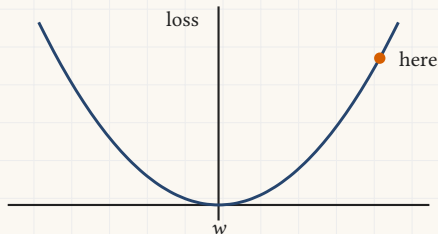
- A) `w = w + alpha * l`
- B) `w = w - alpha * l`
- C) `w = alpha * l`
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A

Move the guess by a step, in the direction the miss points. B walks away from the answer, and D blows

2. The tilt, on the side of the valley

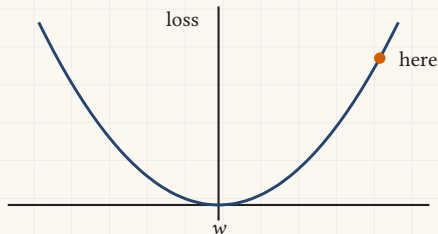
The curve is the loss. The dot is where the guess is now. What is the gradient there?



- A) about -12
- B) exactly 0
- C) about 0.1
- D) about $+12$

2. The tilt, on the side of the valley

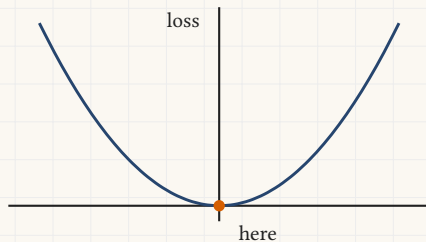
The curve is the loss. The dot is where the guess is now. What is the gradient there?



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D Uphill to the right, so the tilt is positive, and the wall is steep, so it is large. The step the algorithm takes is that number times alpha, and it moves the guess the other way, back down.

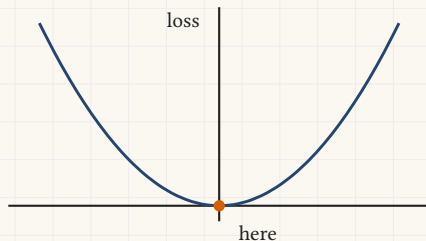
3. The tilt, at the bottom



Same curve. The dot has moved.

- A) about -12
- B) exactly 0
- C) about $+0.5$
- D) about $+12$

3. The tilt, at the bottom



Same curve. The dot has moved.

- A) about -12
- B) exactly 0
- C) about $+0.5$
- D) about $+12$

B Flat ground: the tilt is zero, the step is zero, and the grasshopper stops. Careful, though. It stops on ANY flat spot, lowest or not.

4. Which of these can draw any shape at all?

Given enough of them. More than one answer is right.

- A) Neural networks
- B) Straight lines
- C) Polynomials
- D) Loss functions

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A and C

Both. A network with enough bends can approximate any shape, and so can a polynomial of high enough degree. Neither promise says how many bends or what degree, which is the catch. A straight line can only ever be a straight line, and a loss function is the thing that measures a shape rather than a thing that draws one.

5. The network inside a language model

You type half a sentence and press enter. What does the network itself hand back?

- A) A word
- B) A matrix of words
- C) One probability for every word it knows
- D) The word with the highest score

5. The network inside a language model

You type half a sentence and press enter. What does the network itself hand back?

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C It never hands back a word. It hands back a chance for each of the fifty thousand or so words it knows, and something outside the network picks one. D is what that something does next, and it is where the temperature dial lives.

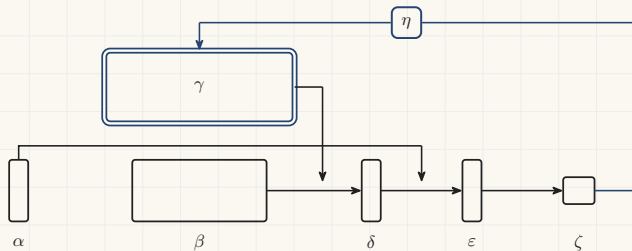
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Section 2

The same five boxes, twice

The picture from one o'clock, with the words taken out

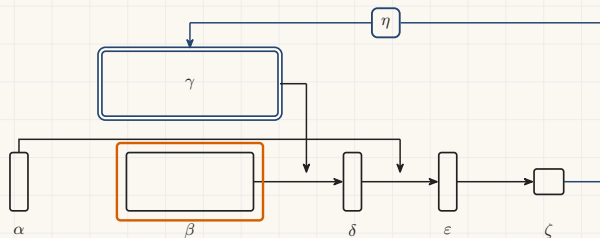
Every application today was this one diagram. The names are gone; the shapes are not.



We are going to fill it in, one box at a time, for a network that prices houses. Then we will do it again for a language model.

6. House prices: the data going in

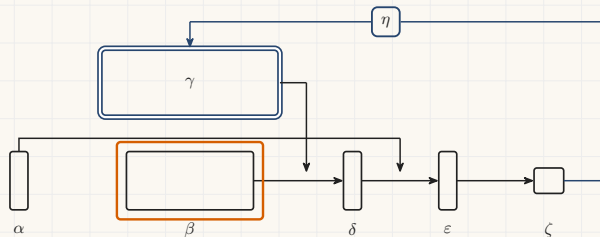
A network that predicts house prices. What is β ?



- A) The prices the network guesses
- B) One row per house, one column per measurement
- C) How wrong each guess was
- D) The knobs of the network

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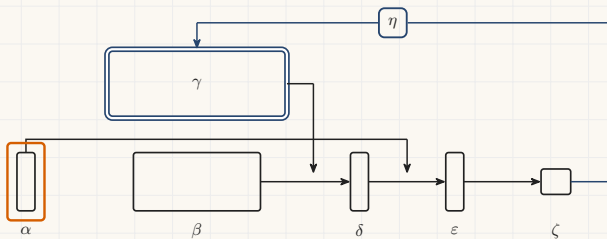


- A) The prices the network guesses
- B) One row per house, one column per measurement
- C) How wrong each guess was
- D) The knobs of the network

B The measurements: size, rooms, age, distance. One row is one house. Everything the network is ever told about a house is in that row.

7. House prices: the box on the left

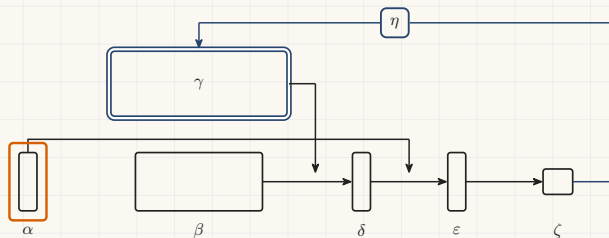
A network that predicts house prices. What is α ?



- A) The measurements of the houses
- B) The prices the network guesses
- C) The prices actually paid
- D) The gradient

7. House prices: the box on the left

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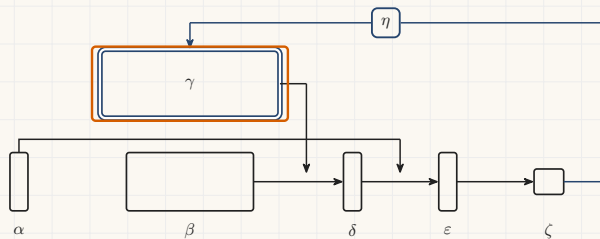


- A) The measurements of the houses
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- D) The gradient

C The truth. What each house really sold for, one number per house. It is the only thing in the picture the network is never shown while it is guessing.

8. House prices: the big box on top

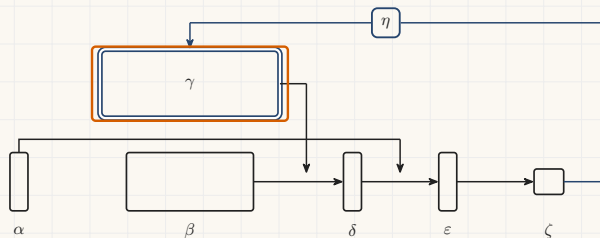
A network that predicts house prices. What is γ ?



- A) The neural network
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- D) The learning rate

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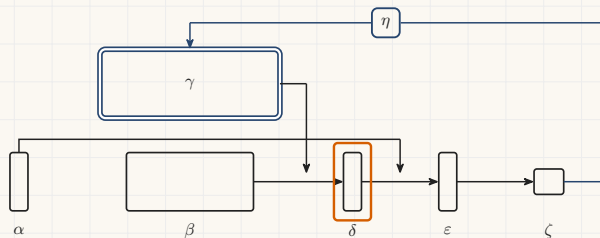


- A) The neural network
- B) The training data
- C) The loss
- D) The learning rate

A The model itself: the weighted sums and the bends, and every knob training is allowed to turn. Today it was linear regression at nine o'clock and a network at eleven, and the box did not move.

9. House prices: the box after the arrow

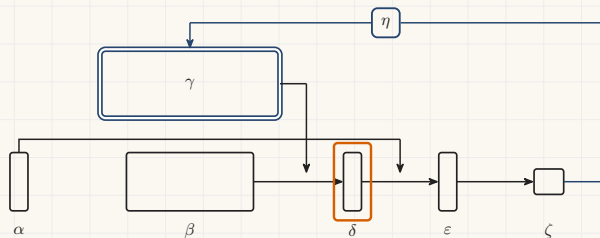
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- A) The prices actually paid
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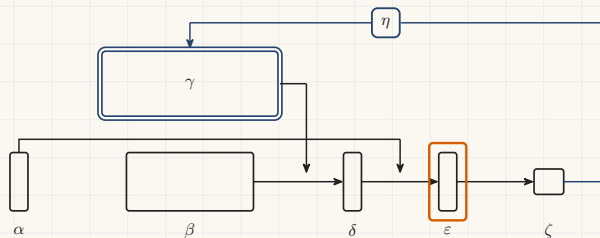


- A) The prices actually paid
- B) The measurements
- C) The number of houses
- D) The prices the network estimates

D The estimate, one per house. In your homework tonight this is the number that came out typically about fifty thousand francs away from the truth.

10. House prices: the box where the two streams meet

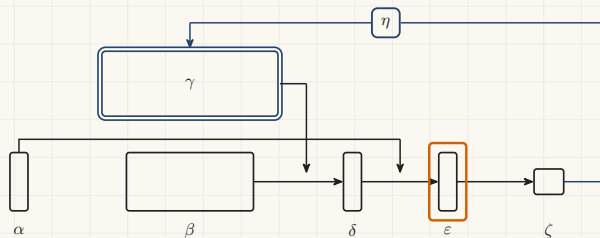
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- A) The average price on the street
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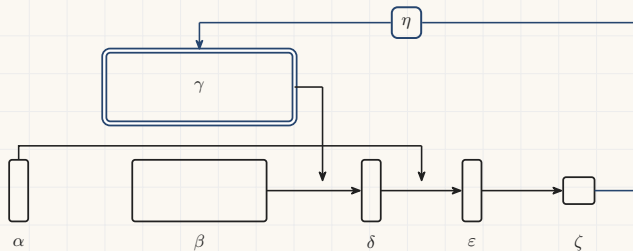


- A) The average price on the street
- B) How wrong each estimate was
- C) The next guess
- D) The number of knobs

B The miss, house by house: the truth and the estimate arriving in the same box. The box after it averages them into one number, and the arrow curving back over the top is the correction.

Now the same picture, for a language model

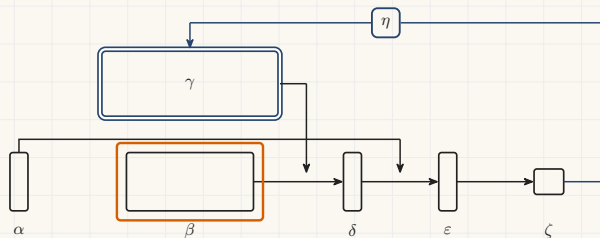
Nothing about the diagram changes. Only what is in the boxes.



One row is a piece of text with its next word cut off. Every position of every document gives you one row, which is how the pile gets big enough to matter.

11. The language model: the data going in

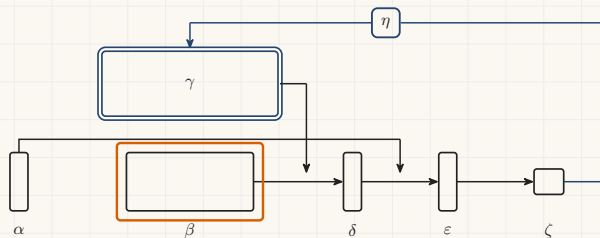
A language model in training. What is β ?



- A) The words so far, each one turned into a list of numbers
- B) The whole internet, as text
- C) The word that comes next
- D) The rules of English grammar

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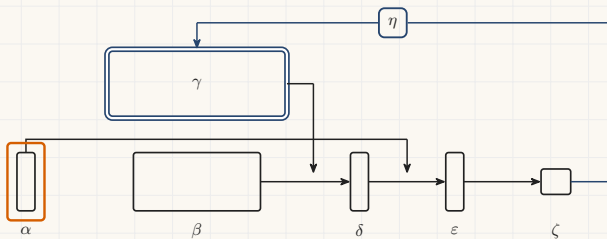
- A) The words so far, each one turned into a list of numbers
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A

The context: the text up to this point, with every word already turned into numbers, because a network only ever eats numbers. Nobody wrote a rule of grammar anywhere in this picture.

12. The language model: the box on the left

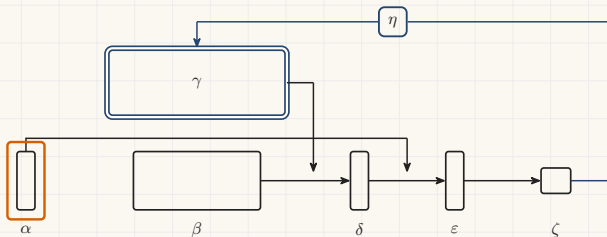
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- A) The answer a human wrote
- B) The question the user typed
- C) The word that actually came next
- D) A score out of ten

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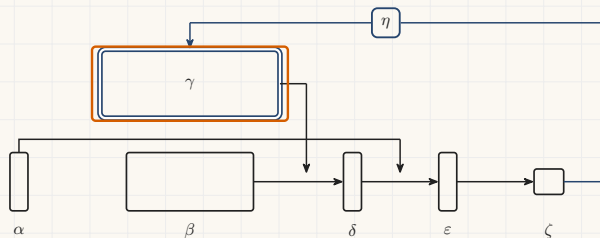


- A) The answer a human wrote
- B) The question the user typed
- C) The word that actually came next
- D) A score out of ten

C The truth, and it is free. Every piece of text in the world already says what came next, so nobody has to label anything. That is why this worked at the size it did.

13. The language model: the big box on top

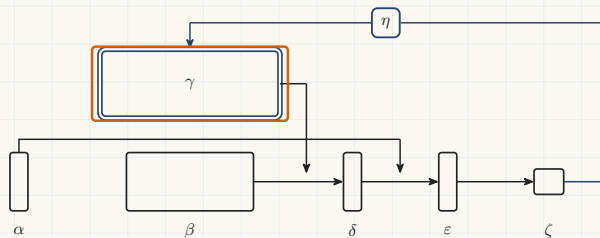
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- C) A database of sentences
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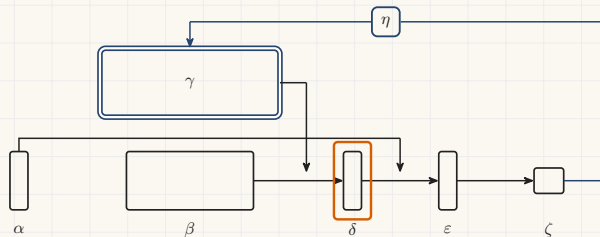


- A) A search engine
- B) A transformer
- C) A database of sentences
- D) A set of rules written by linguists

B A transformer, which is a very large pile of weighted sums and bends. It stores no sentences and looks nothing up. The day never opened this box, and this is the one place it has a name.

14. The language model: the box after the arrow

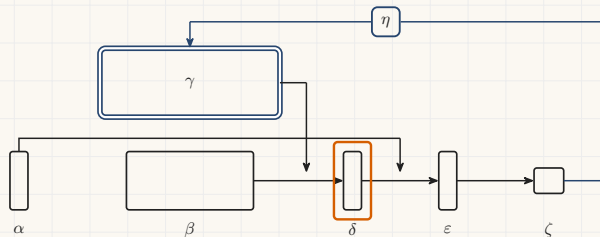
A language model in training. What is δ ?



- A) The next word
- B) The finished answer
- C) A number between zero and one
- D) A matrix: one row per position, and in each row a probability for every word

14. The language model: the box after the arrow

A language model in training. What is δ ?



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- B) The finished answer
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D Careful. Not a word and not a sentence. For every position in the text it gives a chance to every word it knows, so it is a whole table of numbers. Question 5 again, and this is where it lives in the picture.
Foundations of AI, the closing quiz

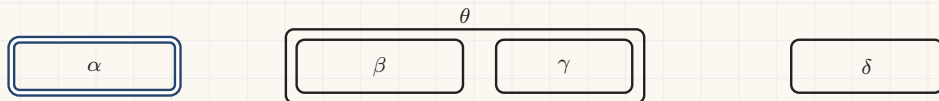
3

Section 3

The agent

The picture from the agentic AI slides

Four actors, and one of them is not a program you wrote. Names taken out.

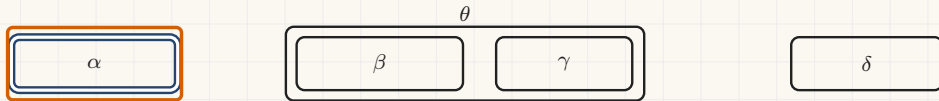


γ is **the**

client, the piece of plumbing that speaks the protocol, and θ is **the host application**, the thing you actually installed. We are not asking about those two. We are asking about the other three, left to right.

15. The agent: the box on the left

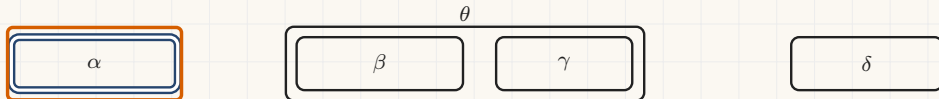
What is α ?



- A) The user
- B) The model
- C) The tool
- D) The internet

15. The agent: the box on the left

What is α ?

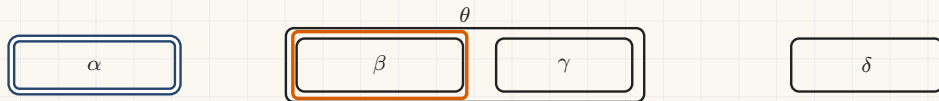


- A) The user
- B) The model
- C) The tool
- D) The internet

B The model, and all it ever does is write text. It runs nothing, opens nothing and sends nothing. Everything to its right exists because of that.

16. The agent: the box next to it

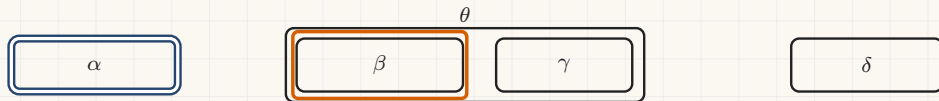
What is β ?



- A) The harness: the ordinary program around the model
- B) A second, larger model that supervises the first
- C) The user's keyboard
- D) The training data

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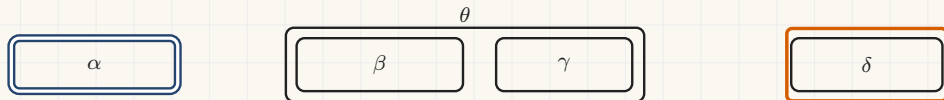
A

The harness. Ordinary code that you can read, which reads what the model wrote, decides whether to allow it, runs it, and hands the result back. This is where you govern an agent, and it is the answer to

Foundations of AI, the closing quiz
question 8 of your homework.

17. The agent: the box on the right

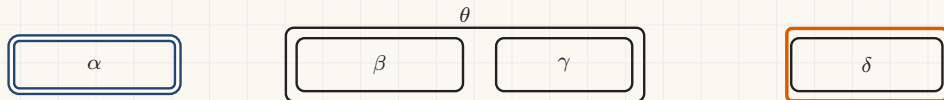
What is δ ?



- A) A backup of the model
- B) The user's laptop
- C) The server, which holds the tools and runs them
- D) The place the answer is stored

17. The agent: the box on the right

What is δ ?



- A) A backup of the model
- B) The user's laptop
- C) The server, which holds the tools and runs them
- D) The place the answer is stored

C The server. It owns the tools, it describes them on request, and it is the thing that actually executes one.

It can be somebody else's, which is the whole point of having a protocol.

18. What happens first?

Before the model has written a single word, one thing has already happened between the client and the server.

- A) The model asks the user what it is allowed to do
- B) The client asks the server which tools it has, and puts the answer into the model's context
- C) The server runs every tool once to check that they work
- D) The harness writes the tool descriptions by hand

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 - B) The client asks the server which tools it has, and puts the answer into the model's context
 - C) The server runs every tool once to check that they work
 - D) The harness writes the tool descriptions by hand
- B** The client asks, the server answers with the tool definitions, and those definitions go into the context. A model cannot call a tool it has never been told about, and that list is how it is told.

19. And the last step of all

The server has run the tool and produced a result. What happens to it?

- A) It is shown to the user and then discarded
- B) It is written into the model's weights
- C) It is sent straight back to the server
- D) It is appended to the transcript, and the model is called again with it

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D Appended, and round again. The model does not remember anything: it is called afresh each time with a longer transcript. That loop, and the checks in the harness, are the whole of what an agent is.

That is the day

- A guess, a measurement of how wrong it was, and a correction. Nine o'clock to five.
- The same five boxes behind a house price and behind a sentence.
- The only honest number comes from data the model never saw.
- A model writes text. A harness does things. Govern the harness.

Your project is on the course page: about an hour, in the browser, and it builds the house price network you met at question 6. Thank you for a Saturday.